

ASSIGNMENT

ON

BASIC STATISTICS & PROBABILITY
(Course code: AED 301)

Topic: Assignment-02 (Problem Solving)

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①

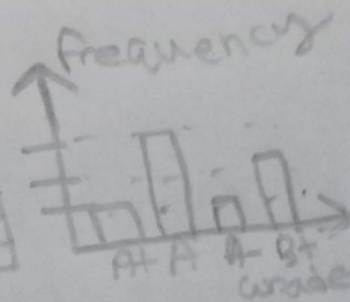
Statistics: Methodology of collecting, organizing, summarizing and analyzing data and drawing conclusion from the data.

Examples: Here's grades of 7 students from 8th batch of Software Engineering (SWE) Department: B+, A, A, A+, A-, B+

Organized data: A+, A, A, A, B+, B+

Summarize data:

Grade	Frequency	%
A+	01	$(1/7 \times 100)\%$
A	03	$(3/7 \times 100)\%$
A-	01	$(1/7 \times 100)\%$
B+	02	$(2/7 \times 100)\%$
	07	



Scope of statistics:

- i) Agricultural Problem: Is new fertilizers more productive?
- ii) Medical Problem: What's right amount of dosage of drug to treatment? Clinical trial
- iii) Economics: What will be the unemployment rate next year?
- iv) Technical Problem: How to improve the Quality of Product?

Limitations: i) Works only with numerical data

ii) Results maybe misleading if data is biased

iii) Shows averages, not individual truth.

iv) Requires Expert Interpretation

v) Can't establish exact cause-effect, only relationships.

②

Variable: Refers to a characteristics or property of an object or individual that can vary from one individual or object to another.

Example: Religion, Age, Height, weight.

- Types of variables:
- Qualitative/categorical Variable associated with non numerical values; Example: Grade, Religion
 - Quantitative Variable associated with numerical observations
 - Discrete Variable that takes any specific values in a given range and is countable. Example: Number of students in a class. Discrete is specifically got value or no value. value is 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100.
 - Continuous Variable that takes theoretically any value in a given range and can be measured. Example: Height, weight, voltage, exam mark.

③

Scale of Measurement: Determines how data is measured and classified.

Scale: A device used to measure any event. Variables are defined and categorized using different Scale of Measurements.

Each level of measurement scale has following properties:-

i) Nominal Scale: Categorizes items. Example: Religion, Blood Group, Gender, Nationality.

ii) Interval Scale: Categorizes and ranks items and has equal intervals with arbitrary zero (fake zero). Example: ① Cup A 10°C , Cup B 20°C (Temperature in $^{\circ}\text{C}$ standard for water), ② Calendar Year (Initial point isn't same)

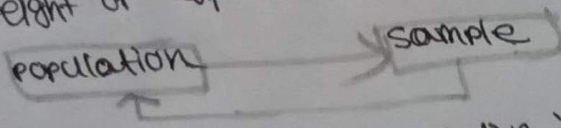
iii) Ordinal Scale: Categorizes and ranks items. Example: Foot rating, Grade in exam, place (Position/rank) in a race.

iv) Ratio Scale: Categorizes and Ranks items and has equal intervals with true-zero. All arithmetic operations are meaningful. Example: ① Cup A 10K Cup B 20K (Temperature in Kelvin) Initial point same for both, ② Height, ③ Weight, ④ Age of person.

④

Population: Collection of all individuals/items under consideration in a statistical study. Example: Height of all students of MV.

Sample: Subject of population, the part from which data is collected. Example: Height of all selected students in a scale.



Conclusion about population is drawn from sample.

Data: Observations that occur in a planned experiment. Raw materials or statistics of scientific investigation.

Age	Variable
20	Data
22	Data
23	Data
25	Data

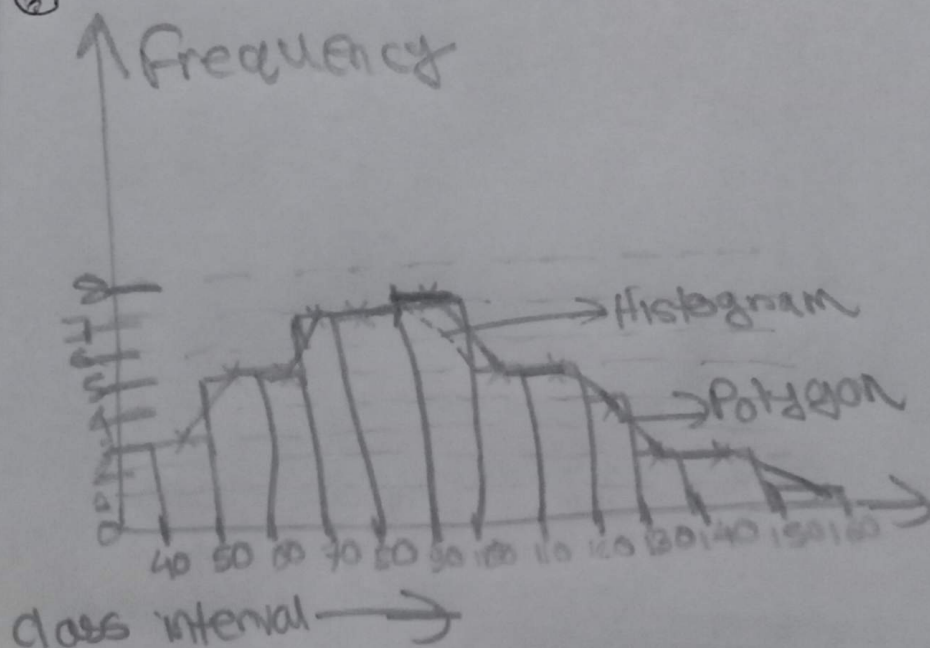
(5)

Recorded time (in minutes) spent by students from various departments studying on weekdays at a university called Blue Ridge University are: 45, 50, 60, 55, 70, 85, 75, 90, 40, 95, 80, 100, 110, 120, 115, 65, 55, 70, 75, 50, 60, 85, 90, 95, 100, 105, 70, 80, 85, 95, 110, 115, 50, 45, 60, 85, 70, 85, 90, 95, 100, 105, 110, 120, 125, 130, 135, 140, 145, 150.

@ Min value = 40, max value = 150, range = $(150 - 40) + 1 = 111$, class interval = 10,
number of classes = $\frac{111}{10} = 11.1 = 12$.

	class interval	inequality	frequency	Mid-point
1	40-50	$40 \leq x < 50$	3	45
2	50-60	$50 \leq x < 60$	5	55
3	60-70	$60 \leq x < 70$	5	65
4	70-80	$70 \leq x < 80$	8	75
5	80-90	$80 \leq x < 90$	6	85
6	90-100	$90 \leq x < 100$	7	95
7	100-110	$100 \leq x < 110$	5	105
8	110-120	$110 \leq x < 120$	5	115
9	120-130	$120 \leq x < 130$	3	125
10	130-140	$130 \leq x < 140$	2	135
11	140-150	$140 \leq x < 150$	2	145
12	150-160	$150 \leq x < 160$	1	155
			total = 50	

(6)



@ From the drawn histogram, we can conclude that mode is around 90-100, and the tail is towards right side which means positive skewness, that is $\text{mean} > \text{median}$. Thus, the data is skewed and the data is right skewed.

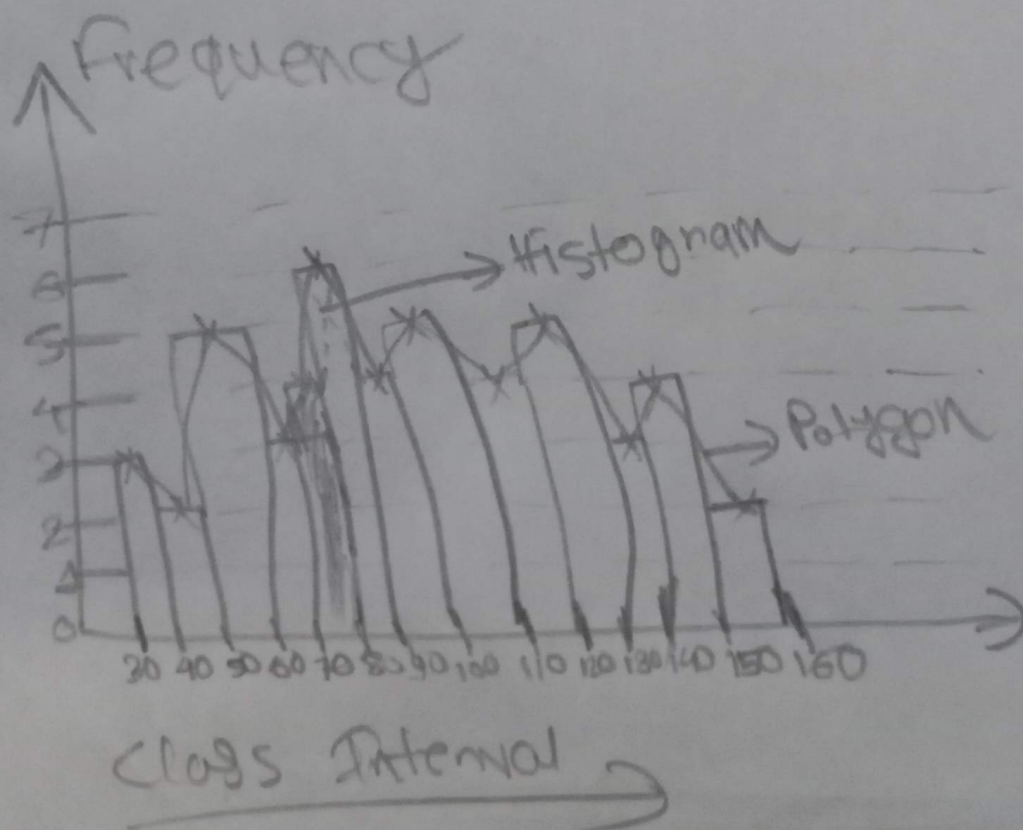
(Ans)

(6)

Recorded weekly exercise duration (in minutes) of residents in a town called Willow Creek are: 30, 45, 60, 50, 75, 90, 80, 95, 95, 100, 85, 110, 120, 125, 115, 55, 65, 75, 85, 40, 50, 70, 80, 90, 100, 105, 115, 120, 130, 140, 145, 150, 50, 25, 55, 60, 70, 80, 85, 95, 100, 105, 110, 120, 125, 130, 140, 145, 150.

① Min-value = 30, max-value = 150, range = $(150 - 30) = 120$, class interval = 10, no of class = $\frac{120}{10} = 12$ or 13.

	class interval	The equality	Frequency	Mid-Point
1	30-40	$30 \leq x < 40$	3	35
2	40-50	$40 \leq x < 50$	2	45
3	50-60	$50 \leq x < 60$	5	55
4	60-70	$60 \leq x < 70$	3	65
5	70-80	$70 \leq x < 80$	4	75
6	80-90	$80 \leq x < 90$	6	85
7	90-100	$90 \leq x < 100$	4	95
8	100-110	$100 \leq x < 110$	5	105
9	110-120	$110 \leq x < 120$	4	115
10	120-130	$120 \leq x < 130$	5	125
11	130-140	$130 \leq x < 140$	3	135
12	140-150	$140 \leq x < 150$	4	145
13	150-160	$150 \leq x < 160$	2	155
		Σ x	Total = 50	



② From the drawn histogram the mode is around 80-90 and the tail is towards the right side, which means positive skewness that is mean > median. Thus, the data is skewed and the data is right-skewed. (Determined)

7

The data (number of sales) made by each of the 10 sales representatives of a sales company in a week are: 12, 15, 20, 17, 14, 18, 16, 22, 19, 21.

From the above data, total number of data, $n=10$.

$$\text{Mean, } \bar{x} = \frac{\sum x_i}{n} = \frac{12+15+20+17+14+18+16+22+19+21}{10}$$

$$\therefore \text{Mean, } \bar{x} = 17.4$$

Arrange data: 12, 14, 15, 16, 17, 18, 19, 20, 21, 22

$$\text{Median} = \frac{n}{2} = \frac{10}{2} = 5^{\text{th}} \text{ term}$$

$$\therefore \text{Median} = \frac{5^{\text{th}} + 6^{\text{th}}}{2} = \frac{17+18}{2} = 17.5$$

Since, no data was repeated so there's no frequency and no mode for the data.

From (a) we get, Mean = 17.4 and Median = 17.5.

Here Median > Mean.

But difference is too small. We conclude Median > Mean. Data is symmetric, no skewness.

Since from (b), Mean > Median, so mean is the suitable central measure. As the value of mean and median is almost equal (too small difference), meaning there's no outlier in the above data set, so we use mean as central measure instead of median.

Ans to que no. 8

Number of books	Frequency	Cumulative Frequency
1	5	5
2	8	13
3	10	23
4	6	29
5	3	32

Table shows numbers of books read by a group of students.

$$\text{Mean, } \bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{(1 \times 5) + (2 \times 8) + (3 \times 10) + (4 \times 6) + (5 \times 3)}{5+8+10+6+3} = 2.8 \approx 3$$

$$\text{Median} = \frac{n}{2} = \frac{32}{2} = 16^{\text{th}} \text{ term}$$

$$\text{Median} = \frac{16+17}{2} = \frac{3+3}{2} = 3$$

$$\text{Mean} = 3, \text{Median} = 3, \text{Mode} = 3 \text{ (because of high frequency)}$$

(Answer)

From (a), Mean = 3, Median = 3, Mode = 3.

Relation is: Mean = Median = Mode

$\therefore$ Data is symmetric, no skewness.

(Answer)

From (b), we conclude data is symmetric, no skewness, so no outlier and so we choose mean as the suitable central measure (since Mean = Median = Mode). Thus, no outlier since values are equal. If there's no outlier, we choose mean as the suitable central measure.

(Answer)

9

The weekly income (in dollars) of 50 employees at a factory is -

Income Range (\$)	Frequency	Cumulative frequency	Mid-point
300-400	5	5	350
400-500	12	17	450
500-600	18	35	550
600-700	10	45	650
700-800	5	50	750

a) Mean $\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{(350 \times 5) + (450 \times 12) + (550 \times 18) + (650 \times 10) + (750 \times 5)}{5 + 12 + 18 + 10 + 5} = 548$ ①

Median $= \frac{n}{2} = \frac{50}{2} = 25^{\text{th}}$ term.

$L = 500, F = 17, f = 18, c = 100, A_1 = 6, A_2 = 8.$

∴ Median $= L + \left(\frac{n/2 - F}{f} \right) \times C = 500 + \left(\frac{50/2 - 17}{18} \right) \times 100 = 544.44$ ②

∴ Mode $= L + \left(\frac{A_1}{A_1 + A_2} \right) \times C = 500 + \left(\frac{6}{6 + 8} \right) \times 100 = 542.85$ ③

④ From ①, Mean = 548, Median = 544.44, Mode = 542.85.

Relation is: Mean > Median > Mode.

∴ Data is positively skewed (Answer)

⑤ From ⑤, we conclude, mean > median, so skewness is positive.

∴ Suitable Central Measure is Median.

Because outliers won't affect median measure. (Answer)

10

The heights (in m) of 12 students of class are: 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205.

① $n = 12$.

First Quartile, $Q_1 = \frac{1}{4} \times 12 = 3^{\text{rd}}$ term.

Second Quartile, $Q_2 = \frac{2}{4} \times 12 = 6^{\text{th}}$ term.

Third Quartile, $Q_3 = \frac{3}{4} \times 12 = 9^{\text{th}}$ term.

$Q_1 = \frac{3^{\text{rd}} + 4^{\text{th}}}{2} = \frac{160 + 165}{2} = 162.5$ ①

$Q_2 = \frac{6^{\text{th}} + 7^{\text{th}}}{2} = \frac{175 + 180}{2} = 177.5$ ②

$Q_3 = \frac{9^{\text{th}} + 10^{\text{th}}}{2} = \frac{190 + 195}{2} = 192.5$ ③

④ From ①, $Q_1 = 162.5, Q_2 = 177.5, Q_3 = 192.5$

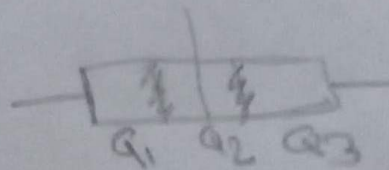
Now, $Q_3 - Q_2 = 192.5 - 177.5 = 15$

$Q_2 - Q_1 = 177.5 - 162.5 = 15.$

$Q_3 - Q_2 = Q_2 - Q_1.$

∴ Data is symmetric, no skewness

(Answer)



11)

The ages of participants in a community workshop are:-

Age (years)	Frequency	Cumulative frequency
20	6	6
25	8	14
30	10	24
35	5	29
40	4	33

① $n=33$

$Q_1 = \frac{1}{4} \times 33 = 8.25 = 9^{\text{th}} \text{ obs. } Q_1 = 25$

$Q_2 = \frac{2}{4} \times 33 = 16.5 = 17^{\text{th}} \text{ obs. } Q_2 = 30$

$Q_3 = \frac{3}{4} \times 33 = 24.75 = 25^{\text{th}} \text{ obs. } Q_3 = 35$

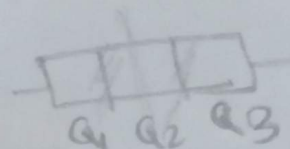
② From ①, we get, $Q_1 = 25, Q_2 = 30, Q_3 = 35$

Now, $Q_3 - Q_2 = 35 - 30 = 5$

$Q_2 - Q_1 = 30 - 25 = 5$

$Q_3 - Q_2 = Q_2 - Q_1$

∴ Data is symmetric, no skewness. (Answer)



12)

The distribution of test scores for 60 students is:-

Score Range	Frequency	Cumulative frequency
40-50	8	8
50-60	20	28
60-70	15	43
70-80	10	53
80-90	5	58

① $n=60$

$Q_1 = \frac{1}{4} \times 60 = 15^{\text{th}} \text{ obs.}$

$L=50, F=8, f=12, C=10$

$Q_1 = L + \left(\frac{\frac{1}{4}n - F}{f} \right) \times C$
 $= 50 + \left(\frac{\frac{1}{4} \times 60 - 8}{12} \right) \times 10$

$Q_1 = 55.83$

$Q_2 = \frac{2}{4} \times 60 = 30^{\text{th}} \text{ obs}$

$L=60, F=20, f=20, C=10$

$Q_2 = L + \left(\frac{\frac{2}{4}n - F}{f} \right) \times C$
 $= 60 + \left(\frac{\frac{2}{4} \times 60 - 20}{20} \right) \times 10$

$Q_2 = 65$

$Q_3 = \frac{3}{4} \times 60 = 45^{\text{th}} \text{ obs}$

$L=70, F=40, f=15, C=10$

$Q_3 = L + \left(\frac{\frac{3}{4}n - F}{f} \right) \times C$
 $= 70 + \left(\frac{\frac{3}{4} \times 60 - 40}{15} \right) \times 10$

$Q_3 = 73.33$

② From ① we get, $Q_1 = 55.83, Q_2 = 65, Q_3 = 73.33$

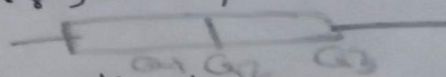
$Q_3 - Q_2 = 73.33 - 65 = 8.33$

$Q_2 - Q_1 = 65 - 55.83 = 9.17$

$Q_3 - Q_2 < Q_2 - Q_1$

∴ Data is negatively skewed.

(Answer)



(13)

The daily rainfall (in mm) recorded over 7 days in a city is: 12, 15, 18, 20, 25, 30, 35.
 Min-value = 12, max-value = 35, Range = 35 - 12 = 23, $n = 7$

$$Q_1 = \frac{1}{4} \times 7 = 1.75 \approx 2^{\text{nd}} \text{ obs.} \quad Q_1 = 15$$

$$Q_3 = \frac{3}{4} \times 7 = 5.25 \approx 6^{\text{th}} \text{ obs.} \quad Q_3 = 30$$

$$IQR = Q_3 - Q_1 = 30 - 15 = 15$$

$$\sum x_i^2 = 3843, \sum x_i = 155$$

$$\begin{aligned} \text{Variance, } S^2 &= \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2 \\ &= \frac{3843}{7} - \left(\frac{155}{7} \right)^2 \\ \therefore S^2 &= 58.69 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Standard variance, } S &= \sqrt{58.69} \\ \therefore S &= 7.66 \text{ mm} \quad (\text{Ans}) \end{aligned}$$

(14)

The number of hours spent on homework by a group of students in a week is: —

Hours	Frequency	Cumulative Frequency
5	4	4
6	6	10
7	8	18
8	5	23
9	2	25

① Min-value = 5, max-value = 9, Range = 9 - 5 = 4, $n = 25$.

$$Q_1 = \frac{1}{4} \times 25 = 6.25 = 7^{\text{th}} \text{ obs.} \quad Q_1 = 6$$

$$Q_3 = \frac{3}{4} \times 25 = 18.75 = 19^{\text{th}} \text{ obs.} \quad Q_3 = 8$$

$$IQR = Q_3 - Q_1 = 8 - 6 = 2$$

$$\sum f_i x_i^2 = 1190, \sum f_i x_i = 170$$

$$\begin{aligned} \text{Variance, } S^2 &= \frac{\sum f_i x_i^2}{n} - \left(\frac{\sum f_i x_i}{n} \right)^2 \\ &= \frac{1190}{25} - \left(\frac{170}{25} \right)^2 \end{aligned}$$

$$\therefore S^2 = 1.36 \text{ hours}^2$$

$$\text{Standard variance, } S = \sqrt{1.36} \Rightarrow S = 1.16 \text{ hours} \quad (\text{Ans})$$

② From ①, $Q_1 = 6, Q_3 = 8$.

Now, $Q_2 = \frac{2}{4} \times 25 = 12.5 \approx 13^{\text{th}} \text{ obs.} \quad Q_2 = 7$.

$$Q_3 - Q_2 = 8 - 7 = 1$$

$$Q_2 - Q_1 = 7 - 6 = 1$$

$Q_3 - Q_2 = Q_2 - Q_1 = 1$ means no skewness. If there's no skewness then standard deviation is the most suitable dispersion measure.

(Answer)

15)

The distribution of weights (in kg) of 50 athletes is:-

Weight Range	Frequency	Cumulative Frequency	Mid-point
50-55	7	7	52.5
55-60	13	20	57.5
60-65	20	40	62.5
65-70	8	48	67.5
70-75	2	50	72.5

① Min-value = 50, Max-value = 75, Range = 75 - 50 = 25, $n = 50$.

$$Q_1 = \frac{1}{4} \times 50 = 12.5 \approx 13^{\text{th}} \text{ obs. } Q_1 = 55 + \left(\frac{14 \times 50 - 7}{13} \right) \times 5 \Rightarrow Q_1 = 57.12$$

$$Q_3 = \frac{3}{4} \times 50 = 37.5 \approx 38^{\text{th}} \text{ obs. } Q_3 = 60 + \left(\frac{14 \times 50 - 20}{20} \right) \times 5 \Rightarrow Q_3 = 64.38$$

$$IQR = Q_3 - Q_1 = 64.38 - 57.12 = 7.26$$

$$\sum f x_i^2 = 187362.5 \quad \sum f x_i = 3050$$

$$\text{variance, } S^2 = \frac{\sum f x_i^2}{\sum f} - \left(\frac{\sum f x_i}{\sum f} \right)^2$$

$$= \frac{187362.5}{50} - \left(\frac{3050}{50} \right)^2$$

$$\therefore S^2 = 26.25 \text{ Kg}^2$$

$$\text{standard deviation, } S = \sqrt{26.25} \Rightarrow S = 5.12 \text{ Kg}$$

② From ① we get $Q_1 = 57.12$, $Q_2 = 64.38$

$$\text{Now } Q_2 = \frac{2}{4} \times 50 = 25^{\text{th}} \text{ obs. } Q_2 = 60 + \left(\frac{214 \times 50 - 20}{20} \right) \times 5 = 61.25$$

$$Q_3 - Q_2 = 64.38 - 61.25 = 3.13$$

$$Q_2 - Q_1 = 61.25 - 57.12 = 4.13$$

$Q_3 - Q_2 < Q_2 - Q_1$ means negative skewness. If

skewness is present in the data, we choose IQR as suitable dispersion measure.

Ans to Q. no. 16

Situations in which the Coefficient of variation (CV) is particularly useful are:

① When units are different

② When the mean of two variables differ significantly, making direct comparison of standard deviations inappropriate.

For Player A:

Score	Frequency
2	2
3	2
4	2
5	2
6	1
7	1
Total	10

$$\text{Mean, } \bar{x} = \frac{\sum f x_i}{\sum f} = \frac{(2 \times 2) + (3 \times 2) + (4 \times 2) + (5 \times 2) + (6 \times 1) + (7 \times 1)}{2 + 2 + 2 + 2 + 1 + 1} = 4.1$$

$$\sum f x_i^2 = 193, \quad \sum f x_i = 41$$

$$\text{standard deviation, } S = \sqrt{\frac{\sum f x_i^2}{\sum f} - \left(\frac{\sum f x_i}{\sum f} \right)^2}$$

$$= \sqrt{\frac{193}{10} - \left(\frac{41}{10} \right)^2}$$

$$\therefore S = 1.58 \text{ goals}$$

(Answer)

16.b-i)

For Player B:

Score	Frequency
1	1
2	1
3	1
4	2
5	1
6	1
7	1
8	1
9	1
10	1

$$\text{Mean, } \bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{(1 \times 1) + (2 \times 1) + (3 \times 1) + (4 \times 2) + (5 \times 1) + (6 \times 1) + (7 \times 1) + (8 \times 1) + (9 \times 1) + (10 \times 1)}{10}$$

$$\therefore \text{Mean, } \bar{x} = 5$$

$$\sum f_i x_i^2 = 320, \sum f_i x_i = 50$$

$$\text{Standard deviation, } S = \sqrt{\frac{\sum f_i x_i^2}{\sum f_i} - \left(\frac{\sum f_i x_i}{\sum f_i}\right)^2}$$

$$= \sqrt{\frac{320}{50} - \left(\frac{50}{10}\right)^2}$$

$$= 2.65 \text{ goals.}$$

16.b-ii) From 16.b-i we get, for Player A, $\bar{x} = 4.1, S = 1.58$

$$CV_A = \frac{S}{\bar{x}} = \frac{1.58}{4.1} = 0.39$$

For Player B, $\bar{x} = 5, S = 2.65$

$$CV_B = \frac{S}{\bar{x}} = \frac{2.65}{5} = 0.53$$

16.b-iii) From 16.b-i, for Player A, $CV_A = 0.39$

for Player B, $CV_B = 0.53$.

$CV_B > CV_A$, which means dispersion of player A is less

than player B. Since, less dispersion = more consistency, so player A has more consistency than player B.

(Ans)